Agentic AI for Robust and Socially Aware Autonomous Driving

R26-DS-017

Project Proposal Report

Bandara W A P S – IT22245960

B.Sc. (Hons) Degree in Information Technology Specialized in Data Science

Department of Computer Science

Sri Lanka Institute of Information Technology Sri Lanka

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Autonomous driving systems have achieved notable progress in perception, planning, and decision-making; however, they still face major challenges in urban environments where pedestrians or cyclists may appear suddenly from occluded areas such as behind parked vehicles, buses, or roadside structures. This limitation reduces the ability of such systems to anticipate risk early and respond safely. To address this issue, this research proposes an Agentic AI-based adaptive intent prediction framework for occluded urban scenarios. The proposed approach aims to predict pedestrian and cyclist crossing intentions before full visual visibility is available by combining partial sensor evidence, contextual environmental cues, and structured reasoning mechanisms. The system integrates multimodal perception, spatio-temporal intent modeling, Chain-of-Thought reasoning, and causal inference to support proactive and socially aware decision-making in autonomous vehicles. The research follows a Design Science Research methodology and evaluates the proposed model using publicly available datasets such as nuScenes and DriveLM together with simulated occlusion scenarios generated in CARLA. Performance will be assessed using metrics including prediction accuracy, early detection capability, robustness under partial visibility, and inference latency. The expected outcome is a more reliable and interpretable intent prediction system that improves early hazard anticipation and enhances the safety of autonomous driving in complex urban environments.

Keywords: Autonomous Driving, Agentic AI, Intent Prediction, Occluded Urban Scenarios, Pedestrian and Cyclist Behavior, Multimodal Perception, Chain-of-Thought Reasoning, Causal Inference, Early Hazard Detection

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
AV	Autonomous Vehicle
AD	Autonomous Driving
LLM	Large Language Model
VLA	Vision-Language-Action
CoT	Chain-of-Thought
XAI	Explainable Artificial Intelligence
CNN	Convolutional Neural Network
RNN	Recurrent Neural Network
LSTM	Long Short-Term Memory
LoRA	Low-Rank Adaptation
DSR	Design Science Research
mAP	Mean Average Precision
QA	Question Answering

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1. INTRODUCTION

1.1 Background and Context

Urban road environments are highly dynamic and unpredictable. Pedestrians often interact closely with vehicles, especially in crowded city areas, near crosswalks, bus stops, or parking zones. In many real-world situations, pedestrians may emerge suddenly from behind parked vehicles, roadside infrastructure, or other obstacles. These occlusion scenarios reduce the visibility available to vehicle sensors, making it difficult for autonomous systems to detect and interpret human intentions early. As a result, vehicles may not have sufficient time to react, which can increase the risk of accidents.

Road safety statistics highlight the importance of addressing this issue. According to the World Health Organization, approximately 1.19 million people die each year due to road traffic accidents worldwide, and a significant proportion of these fatalities involve vulnerable road users such as pedestrians and cyclists. Urban environments in particular present complex interactions between vehicles and humans, making it essential for autonomous systems to understand human behavior and predict potential risks before they occur.

Several existing research studies have explored pedestrian intention prediction and trajectory forecasting to improve autonomous driving safety. For example, spatio-temporal attention-based models have been proposed to analyze pedestrian movement patterns and predict future trajectories in autonomous driving scenarios [1]. Other studies focus on recognizing pedestrian actions and intentions using visual cues such as body posture and motion patterns [2]. Additionally, research on pedestrian–vehicle interaction modeling has attempted to incorporate contextual information from both pedestrians and vehicles to improve crossing intention prediction [3]. While these approaches have demonstrated promising results under normal conditions, many of them rely heavily on clear visual observations and historical trajectory data.To address this limitation, this research proposes an adaptive intent prediction approach for occluded urban scenarios using an Agentic AI framework. The proposed solution aims to infer possible pedestrian intentions even before full visibility by analyzing partial sensor evidence and contextual environmental cues. By combining perception with reasoning mechanisms, the system can anticipate potential crossing behaviors earlier and support safer driving decisions.

As a result, current methods often struggle in **occlusion scenarios**, where pedestrians are partially hidden or not yet visible to the perception system. This limitation reduces the ability of autonomous vehicles to anticipate potential risks early enough. Addressing this gap requires more advanced approaches that can reason about possible human actions even when complete visual information is not available.

To overcome these challenges, this research proposes an adaptive intent prediction approach for occluded urban environments using an Agentic AI framework. The proposed solution aims to utilize partial sensor observations, contextual environmental information, and reasoning mechanisms to infer potential pedestrian intentions before full visibility. By integrating perception, prediction, and reasoning capabilities, the system can support more proactive and socially aware decision-making for autonomous vehicles operating in complex urban environments.

This research is also aligned with global sustainability goals, particularly Sustainable Development Goal 11, which focuses on creating safer and more sustainable urban transportation systems. Furthermore, the study contributes to the **Artificial Intelligence and Intelligent Transportation Systems research cluster**, as it aims to enhance the robustness and safety of autonomous driving technologies through advanced AI-based prediction and reasoning techniques.

1.2 Literature Review

Autonomous driving has become one of the most actively researched areas in modern intelligent transportation systems. Recent studies have focused on improving perception, decision-making, and planning capabilities of autonomous vehicles using advanced artificial intelligence techniques such as deep learning, reinforcement learning, and large language models (LLMs). Despite these advancements, several challenges remain in understanding complex road environments and predicting the behavior of surrounding road users such as pedestrians and cyclists.

Existing Research Approaches

Early autonomous driving systems mainly relied on rule-based algorithms and modular architectures where perception, prediction, and planning were handled as separate modules. While these systems provided a structured framework, they often struggled to handle unpredictable real-world situations. To overcome these limitations, researchers introduced end-to-end deep learning approaches that learn driving behavior directly from large datasets. However, such models often behave like "black boxes," making it difficult to interpret their decisions and guarantee safety in complex traffic environments [2].

Recent research has explored the use of Large Language Models (LLMs) to improve reasoning and decision-making capabilities in autonomous driving systems. For example, Vilho et al. proposed an LLM-based framework called **LLMDriver**, which integrates a language model with driving sensors and memory modules to generate human-like driving decisions. The system retrieves previous driving experiences from a memory database and uses reasoning techniques such as Chain-of-Thought to determine appropriate driving actions. Their experiments using the **nuScenes dataset** and **CARLA simulation environment** demonstrated improved motion planning and collision avoidance compared to traditional methods [1]. However, this work mainly focuses on trajectory planning and collaboration between vehicles rather than predicting the intentions of hidden road users.

Another notable approach is the work by Chen et al., which integrates object-level vector representations with large language models to create an explainable autonomous driving system. In this research, vectorized scene information such as vehicle positions, pedestrian locations, and traffic signals are converted into textual descriptions that can be interpreted by the language model. The authors also introduced a large Driving Question-Answering (DQA) dataset containing approximately 160,000 question-answer pairs generated from 10,000 simulated driving scenarios. Their model demonstrates strong capabilities in understanding driving contexts and generating explainable driving decisions [2]. Despite these

contributions, the system still depends on fully visible scene information and does not explicitly address situations where pedestrians or cyclists appear from occluded areas.

Existing Product and Services

Several other studies have focused on pedestrian intention recognition and trajectory prediction using deep learning techniques. These systems often rely on computer vision models such as convolutional neural networks (CNNs), recurrent neural networks (RNNs), or spatio-temporal attention mechanisms to analyze pedestrian motion patterns. While these models can successfully predict crossing behavior in many cases, they generally require clear visual observations of pedestrians. In real urban environments, pedestrians frequently emerge from partially hidden locations such as behind parked vehicles, buses, or roadside obstacles. In such cases, conventional perception models may fail to detect them early enough to make safe driving decisions.

Existing System Limitation

Existing commercial autonomous driving platforms, including systems developed by companies such as Waymo and Tesla, utilize advanced sensor fusion techniques combining cameras, LiDAR, radar, and GPS data to perceive the environment. Although these systems provide high levels of perception accuracy, they still face challenges when dealing with occluded environments and unexpected human behaviors. Predicting the intention of a pedestrian who is not fully visible remains a difficult problem, especially in dense urban traffic scenarios.

Another limitation of many current systems is the lack of **social awareness** in autonomous driving decisions. Human drivers often anticipate potential hazards by reasoning about indirect cues such as movement behind parked vehicles, sudden cyclist behavior, or crowded pedestrian areas. Traditional AI-based systems do not always capture these subtle contextual signals effectively.

Therefore, while existing research has made significant progress in autonomous driving perception, planning, and explainable decision-making, there remains a clear research gap in **adaptive intention prediction in occluded urban scenarios**. Most existing models rely heavily on complete sensor visibility and fail to predict the behavior of hidden or partially visible road users. Addressing this limitation is critical for improving the safety and reliability of autonomous driving systems.

The proposed research aims to address this gap by developing an **Agentic AI-based intent prediction model** that can infer potential pedestrian or cyclist behavior using partial sensor observations, contextual reasoning, and causal inference techniques. By incorporating reasoning capabilities inspired by recent LLM-based approaches, the proposed system aims to enhance the safety and robustness of autonomous vehicles operating in complex urban environments.

1.3 Research Gap

Although autonomous driving technologies have improved significantly with the help of artificial intelligence and advanced sensor systems, an important capability is still missing in many existing solutions. Most current autonomous systems focus on detecting and predicting the movement of road users only after they become clearly visible to the vehicle's perception system. However, in real urban environments, pedestrians and cyclists often appear suddenly from partially hidden areas such as behind parked vehicles, buses, or roadside structures. In such situations, the system may not have enough time to safely react. Therefore, a key missing capability in many existing approaches is the ability to anticipate the intentions of human agents before they are fully visible to the autonomous vehicle.

Exclusion in Current Research

Many existing research approaches rely on computer vision models, trajectory prediction algorithms, or deep learning techniques that analyze the past motion of detected pedestrians and vehicles. While these models perform well when the surrounding agents are fully observable, they struggle when only partial or indirect information is available. Similarly, recent research exploring large language models for autonomous driving focuses mainly on planning and decision-making based on already detected objects in the environment. These systems assume that perception modules have already provided complete and reliable information about surrounding agents. As a result, they lack the capability to reason about potential risks that originate from occluded or partially visible road users. This limitation can reduce the overall safety of autonomous vehicles operating in complex urban traffic conditions.

Novelty and Creativity

To address these limitations, this research proposes a novel approach that integrates Agentic AI with adaptive intent prediction mechanisms. Unlike conventional systems that rely only on direct visual detection, the proposed approach aims to infer the possible intentions of pedestrians or cyclists using partial sensor evidence, contextual cues, and reasoning capabilities. By introducing a reasoning-driven prediction model, the system can anticipate potential human actions even when the agent is not fully visible. This proactive prediction capability represents a creative advancement compared to traditional reactive systems and contributes to improving the robustness and social awareness of autonomous driving systems.

Evidence-Based Argument

Evidence from recent research supports the need for such improvements in autonomous driving systems. For example, studies exploring LLM-based driving frameworks have demonstrated improved reasoning and planning capabilities, but they mainly focus on trajectory generation and collaborative driving strategies rather than early intent prediction [1]. Similarly, research on explainable autonomous driving systems that combine object-level scene representations with large language models has improved interpretability and contextual understanding of driving scenarios [2]. However, these systems still rely on structured scene information generated from clearly detected objects and do not specifically address situations where pedestrians or cyclists emerge from occluded spaces. Therefore, existing literature highlights a clear research opportunity to develop more adaptive and context-aware prediction models capable of handling occluded urban scenarios.

Existing Research	Approach	Key Limitation	Identified Gap
Explainable AI (XAI) in Autonomous Driving	Provides interpretability using attention maps and language explanations	Improves explainability but not predictive anticipation	Missing adaptive, reasoning-driven early hazard detection mechanism
Driving with LLMs (Chen et al., 2024)	Fuses object-level vector modality with LLM for explainable driving and QA reasoning	Requires structured scene data from fully detected objects	Lack of reasoning under partial visibility or hidden agent scenarios
Spatio-Temporal Pedestrian Intention Prediction Models	Uses CNN, RNN, or attention-based models to predict pedestrian crossing behavior	Depends on clear historical motion data and visual detection	Unable to infer intent before pedestrian becomes visible
Sensor-Fusion Based Autonomous Driving Systems (Industry: Waymo, Tesla)	Combines camera, LiDAR, radar for perception and object tracking	Reactive system – decisions triggered after object detection	Missing proactive prediction based on contextual and causal reasoning
End-to-End Deep Learning Driving Models	Learns driving actions directly from raw sensor input	Acts as a black-box with limited interpretability	Lack of socially-aware reasoning and anticipation of hidden human behavior

Table 1: Research gap comparison

2. OBJECTIVE

2.1 Main Objective

The main objective of this research is to design, develop, and evaluate an **Agentic AI-based adaptive intent prediction model** capable of anticipating pedestrian and cyclist crossing intentions in occluded urban environments using partial sensor observations and contextual reasoning, with the aim of improving early risk detection accuracy compared to conventional visibility-dependent prediction models.

This objective is measurable through experimental evaluation using publicly available autonomous driving datasets and simulated urban scenarios. The proposed model will be assessed based on prediction accuracy, early detection capability, and reduction in reaction delay under occlusion conditions. The system performance will be compared against baseline deep learning models that rely solely on fully visible pedestrian observations.

By focusing on predictive performance under partial visibility scenarios, this research aims to demonstrate measurable improvements in early intent inference and proactive decision support for autonomous vehicles operating in complex urban environments.

2.2 Specific Objectives

To achieve the main objective of this research, the following specific objectives are defined:

1. **To analyze and evaluate existing pedestrian intention prediction models** in order to identify their performance limitations under occluded urban scenarios using benchmark autonomous driving datasets.
2. **To design an occlusion-aware perception and feature extraction framework** that captures partial sensor observations and contextual environmental cues, including road layout, surrounding vehicles, and pedestrian proximity indicators.
3. **To develop an Agentic AI-based adaptive intent prediction model** that integrates spatio-temporal learning with contextual reasoning mechanisms to infer pedestrian or cyclist intentions before full visual detection.
4. **To implement and train the proposed model using publicly available autonomous driving datasets** such as nuScenes, DriveLM and simulate occlusion scenarios to evaluate early-stage prediction capability.
5. **To compare the proposed model against baseline deep learning models** based on prediction accuracy, early detection time, and robustness under partial visibility conditions.
6. **To assess the impact of the proposed system on proactive risk mitigation** by analyzing improvements in early hazard anticipation and decision support in urban driving simulations.

3. METHODOLOGY

This research adopts a **Design Science Research (DSR)** methodology, complemented by controlled experimental evaluation, to develop and validate an intelligent reasoning artifact for adaptive intent prediction in occluded urban environments. The primary objective of this methodology is to design, implement, and rigorously evaluate an **Agentic AI-based Reasoning Engine** capable of anticipating pedestrian and cyclist intentions under partial observability conditions. The methodological structure ensures that both the artifact design and empirical validation are systematically addressed.

3.1 Research Approach

The study follows the principles of **Design Science Research**, which emphasizes the creation and evaluation of innovative artifacts to solve identified real-world problems. In this context, the artifact is a **Vision-Language-Action (VLA) based Reasoning Engine** that integrates perception, contextual reasoning, and predictive modeling to address the limitations of visibility-dependent autonomous driving systems.

The DSR process will involve the following phases:

1. **Problem Identification:**
   Existing autonomous driving systems exhibit limited capability in predicting pedestrian intent in occluded scenarios, where agents emerge from blind spots such as behind parked vehicles or roadside structures.

2. **Objective Definition:**
   The objective is to design an AI artifact that infers intent using partial visual evidence and contextual cues before full visibility is achieved.

3. **Artifact Design and Development:**
   The Reasoning Engine will combine visual perception modules with a structured reasoning framework that incorporates Chain-of-Thought (CoT) and causal inference mechanisms.

4. **Demonstration and Evaluation:**
   The developed model will be evaluated using real-world datasets and simulated high-risk occlusion scenarios to measure predictive performance and reasoning robustness.

This approach ensures that the research contributes not only algorithmic improvements but also a validated AI artifact capable of addressing the identified research gap.

3.2 Technical Methods and Algorithms

The proposed system adopts a multimodal reasoning-based architecture that integrates perception, contextual understanding, and predictive reasoning to anticipate pedestrian intent in occluded urban environments. The framework consists of four primary analytical components:

Detection / Reasoning Method	**Description**	**Algorithms / Techniques**	**Purpose**
Vision-Based Occlusion Detection	Extracts partial visual evidence such as shadows, motion cues, and object boundaries from camera input to detect possible hidden agents.	CNN-based Feature Extraction, Optical Flow Analysis, Edge Detection	Identify partial or indirect visual signals of pedestrians in occluded areas.
Spatio-Temporal Intent Modeling	Models pedestrian motion patterns and temporal behavior using historical and contextual data.	LSTM, Temporal Transformers, Attention Mechanisms	Predict crossing intention based on motion trends and temporal patterns.
Chain-of-Thought (CoT) Reasoning Module	Generates structured reasoning steps linking partial evidence and contextual cues before making intent predictions.	Vision-Language-Action Architecture, Prompt-based Reasoning, Structured Output Generation	Improve interpretability and enable logical decision-making under uncertainty.
Causal Inference-Based Risk Estimation	Models probabilistic relationships between contextual factors (e.g., school zones, crosswalks) and pedestrian behavior.	Bayesian Networks, Causal Graph Modeling, Probabilistic Inference	Infer likelihood of hidden pedestrian emergence in blind-spot scenarios.
Efficient Model Adaptation	Fine-tunes foundation models for domain-specific occlusion scenarios without heavy retraining.	Low-Rank Adaptation (LoRA), Parameter-Efficient Fine-Tuning	Reduce computational cost while maintaining real-time performance.

Table 2 : technical methods & algorithms

3.3 Tools and Platforms

The proposed Agentic AI-based intent prediction system will be developed and evaluated using a combination of modern artificial intelligence frameworks, simulation platforms, and multimodal datasets commonly utilized in autonomous driving research.

Category	Tools / Technologies	Purpose / Justification
Programming Environment	Python	Selected due to its extensive ecosystem for machine learning, computer vision, and AI research, as well as strong community support and rapid prototyping capabilities.
Machine Learning Frameworks	PyTorch, Hugging Face Transformers	Used to implement deep learning architectures, Vision-Language-Action models, and fine-tuning mechanisms such as LoRA. Provides flexibility for multimodal model development.
Foundation Model	Phi-3 Vision (SLM)	Chosen for its low latency, efficient parameter size, and suitability for edge computing environments in autonomous vehicles while maintaining multimodal reasoning capability.
Simulation Platform	CARLA Simulator	Enables generation of controlled urban driving scenarios, including high-risk occlusion edge cases, to evaluate predictive robustness under partial visibility conditions.
Datasets	nuScenes, DriveLM	nuScenes provides real-world multimodal urban driving data, while DriveLM offers structured contextual driving scenarios for reasoning-based model training.
Data Processing & Visualization	NumPy, Pandas, Matplotlib	Used for preprocessing sensor data, feature extraction, statistical analysis, and visualization of model performance metrics.

Model Optimization	Low-Rank Adaptation (LoRA)	Enables parameter-efficient fine-tuning of the foundation model to adapt it for occlusion-aware intent prediction without excessive computational overhead.

Table 3 : tools & platform

3.4 Data Collection and Ethical Considerations

Category	Description	Implementation Details	Purpose
Real-World Dataset Collection	Use publicly available autonomous driving datasets	nuScenes (multimodal urban driving data), DriveLM (contextual reasoning dataset)	Train and evaluate intent prediction under realistic urban conditions
Synthetic Data Generation	Generate controlled occlusion edge cases	CARLA Simulator with configured blind spots, parked vehicles, pedestrian emergence scenarios	Test model robustness in high-risk, rare occlusion scenarios
Data Preprocessing	Extract relevant features from multimodal data	Motion cues, shadow detection, contextual indicators (school zones, crosswalks), object bounding boxes	Prepare structured inputs for reasoning and prediction modules
Data Anonymization	Ensure privacy compliance	Use pre-anonymized datasets; blurred faces and license plates	Protect personal identity and comply with ethical standards
Ethical Compliance	Adhere to institutional and AI safety guidelines	Follow IRB requirements, dataset licensing terms, safety-first AI design principles	Maintain responsible and transparent AI research practices

Table 4 : Data Collection and Ethical Considerations

3.5 Validation Strategy and Performance Metrics

Validation Aspect	Method	Metrics / Evaluation Criteria	Purpose
Quantitative Evaluation (Visible Scenarios)	Benchmark against baseline models	Mean Average Precision (mAP), Accuracy	Measure standard intent prediction performance
Quantitative Evaluation (Occluded Scenarios)	Test under simulated blind-spot conditions	Prediction Horizon, False Positive Rate, False Negative Rate	Evaluate early-stage intent inference capability
Real-Time Performance Testing	Measure inference efficiency	Latency (milliseconds)	Ensure suitability for edge deployment in autonomous vehicles
Comparative Model Analysis	Compare proposed model with traditional visibility-dependent models	Performance improvement percentage, robustness analysis	Demonstrate measurable advantage of proposed approach
Qualitative Human-in-the-Loop Validation	Expert review of Chain-of-Thought reasoning outputs	Logical consistency, interpretability, causal reasoning plausibility	Validate reasoning transparency and social awareness

Table 5 : Validation Strategy and Performance Metrics

4. HIGH-LEVEL SYSTEM ARCHITECTURE

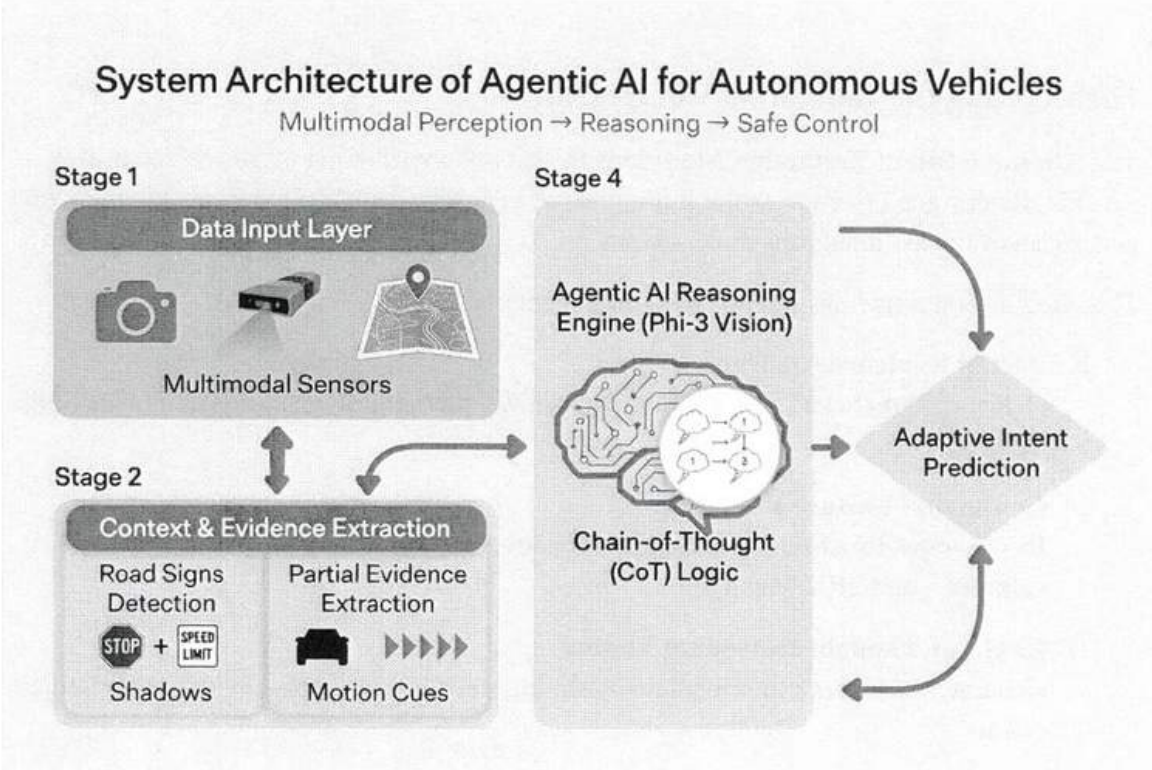


Figure 1 : High level Architecture

The proposed system adopts an **Agentic AI-based modular architecture** designed to enhance robustness and social awareness in autonomous driving environments. The architecture integrates perception, reasoning, prediction, and decision-support components into a cohesive framework capable of operating under partial visibility and occlusion scenarios.

At a high level, the system consists of the following major components:

1. **Perception Module** – Processes multimodal sensor inputs (camera, LiDAR, contextual metadata).
2. **Scene Representation Layer** – Converts raw sensory data into structured object-level representations.
3. **Adaptive Intent Prediction Module (Individual Component)** – Performs occlusion-aware reasoning and predicts pedestrian intent.
4. **Causal Reasoning Engine** – Infers hidden risks using contextual and environmental cues.
5. **Decision Support Interface** – Provides predictive signals to downstream motion planning systems.

The architecture is designed to support dynamic reasoning and early-stage risk anticipation rather than purely reactive detection.

Highlighting the Individual Component

The **Adaptive Intent Prediction Module** is the central contribution of this research. It is specifically designed to address the limitations of visibility-dependent systems by predicting pedestrian or cyclist intentions under occlusion.

This module contains four internal sub-components:

1. **Partial Evidence Analyzer**
   Detects indirect cues such as motion shadows, partial limb visibility, and object-edge disturbances.
2. **Contextual Feature Integrator**
   Incorporates Road semantics such as school zones, pedestrian crossings, parked vehicles, and traffic density.
3. **Chain-of-Thought Reasoning Engine**
   Generates structured intermediate reasoning steps linking partial evidence to predicted intent.
4. **Causal Inference Layer**
   Updates probabilistic belief states to infer hidden agent emergence likelihood in blind-spot regions.

By combining these elements, the component transitions from reactive detection to proactive anticipation.

Component Interactions

The interaction between system components follows a structured flow:

- The **Perception Module** identifies visible objects and potential occlusion regions.
- The **Scene Representation Layer** transforms this information into structured inputs.
- The **Adaptive Intent Prediction Module** receives both direct observations and contextual metadata.
- The **Causal Reasoning Engine** evaluates uncertainty under partial visibility.
- The final intent prediction is forwarded to the **Decision Support Interface**, which informs trajectory planning.

Each component operates independently but shares structured data representations to maintain modularity and scalability.

Data Flow Explanation

The data flow in the system progresses through multiple abstraction layers:

1. **Raw Data Layer**
   Camera frames and contextual metadata are collected.
2. **Feature Extraction Layer**
   Visual embeddings, motion vectors, and semantic annotations are generated.
3. **Reasoning Layer**
   The Vision-Language-Action model processes structured inputs and generates intermediate reasoning outputs.
4. **Prediction Layer**
   The system outputs intent classification along with confidence scores and prediction horizon estimates.
5. **Decision Integration Layer**
   Predictions are transmitted to the motion planning subsystem for proactive maneuver adjustment.

This layered approach ensures transparency and traceability across the inference pipeline.

Scalability and Security Considerations

Scalability

The architecture is designed to scale in three dimensions:

- Model Scalability: Use of Low-Rank Adaptation (LoRA) allows efficient fine-tuning without full model retraining.
- Computational Scalability: Deployment-ready Small Language Models (e.g., Phi-3 Vision) ensure low-latency inference.
- Scenario Scalability: CARLA-generated synthetic data enables expansion to diverse urban environments.

The modular design allows additional reasoning components to be integrated without restructuring the entire system.

Security Considerations

Given the safety-critical nature of autonomous driving systems, security and reliability are prioritized:

- Structured reasoning outputs enhance transparency and auditability.
- Conservative prediction thresholds reduce unsafe overconfidence.
- The system avoids over-reliance on a single sensor modality, reducing vulnerability to adversarial attacks.
- Model integrity checks and parameter validation ensure secure deployment.

Furthermore, the reasoning-based architecture provides an additional layer of interpretability compared to purely end-to-end neural networks.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

The proposed Agentic AI-based intent prediction system involves multiple stakeholders who are directly or indirectly affected by its development and deployment.

1. Autonomous Vehicle Manufacturers
   These stakeholders integrate AI modules into vehicle platforms. They require reliable, low-latency intent prediction systems that improve safety without excessive computational overhead.

2. Autonomous Driving Software Engineers and Researchers
   Developers and researchers are responsible for designing, training, and validating the predictive models. They require interpretable reasoning outputs, modular integration capability, and reproducible performance metrics.

3. Urban Road Users (Pedestrians and Cyclists)
   Although not direct users of the system, they are primary beneficiaries. The system must ensure safer interaction between vehicles and vulnerable road users.

4. Regulatory and Safety Authorities
   Government bodies and transport safety regulators require transparent, auditable AI systems that comply with safety and ethical standards.

5. Fleet Operators and Mobility Service Providers
   Organizations deploying autonomous fleets require systems that enhance operational safety and reduce accident-related liabilities.

6. Academic and Research Institutions
   These stakeholders are interested in extending research on Agentic AI and socially-aware autonomous driving frameworks.

5.2.Functional requirement

The system shall satisfy the following functional requirements:

1. The system shall detect and identify partially visible or occluded pedestrian regions using multimodal sensor inputs.

2. The system shall analyze contextual environmental information, including road type, pedestrian crossings, school zones, and parked vehicle presence.

3. The system shall generate structured Chain-of-Thought reasoning outputs linking partial evidence to predicted intent.

4. The system shall predict pedestrian or cyclist crossing intention before full visual visibility is achieved.

5. The system shall provide confidence scores for each predicted intent outcome.

6. The system shall update prediction results dynamically as new sensor data becomes available.
7. The system shall integrate seamlessly with downstream motion planning modules to support proactive decision-making.
8. The system shall log prediction outputs and reasoning steps for validation and audit purposes.

5.3.Non-functional requirement

1. Performance Requirements

- The system shall maintain real-time inference latency below a predefined threshold (e.g., <100 milliseconds).
- The model shall achieve competitive Mean Average Precision (mAP) under occluded scenarios.
- The system shall support scalable deployment across diverse urban environments.

2. Security Requirements

- The system shall ensure secure handling of sensor data and prevent unauthorized data access.
- The model parameters shall be protected against tampering during deployment.
- The architecture shall minimize vulnerability to adversarial inputs or spoofed sensor signals.

3. Reliability Requirements

- The system shall maintain stable performance under varying lighting and weather conditions.
- The model shall produce conservative risk estimations under uncertainty to avoid unsafe predictions.

4. Usability and Interpretability Requirements

- The system shall provide interpretable reasoning outputs for developer inspection.
- The prediction outputs shall be easily understandable by system engineers and safety auditors.
- The reasoning process shall support explainability to facilitate regulatory compliance.

5. Scalability Requirements

- The architecture shall support parameter-efficient fine-tuning using techniques such as LoRA.
- The system shall be adaptable to additional sensor modalities without significant redesign.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The primary target market for the proposed Agentic AI-based intent prediction system consists of **autonomous vehicle manufacturers and autonomous driving technology companies** seeking to enhance safety and robustness in urban driving environments. This includes organizations developing Level 3–Level 5 autonomous driving platforms.

A secondary target market includes:

- Mobility-as-a-Service (MaaS) providers deploying autonomous fleets
- Advanced Driver Assistance System (ADAS) developers
- Smart city transportation solution providers
- Automotive AI software vendors

Customer Persona

The ideal customer persona can be described as:

- A **Chief Technology Officer (CTO)** or **AI Systems Architect** within an autonomous vehicle company.
- Responsible for improving system safety, regulatory compliance, and real-time decision reliability.
- Seeking AI modules that enhance predictive performance under complex urban scenarios.
- Interested in scalable, interpretable, and deployable edge-computing solutions.

These customers prioritize reliability, regulatory alignment, computational efficiency, and measurable safety improvements.

6.2 Value Proposition

The proposed system offers a clear value proposition:

1. **Early Risk Detection Under Occlusion**
   Unlike traditional systems that react after pedestrian detection, the proposed model anticipates potential intent before full visibility.
2. **Improved Urban Safety**
   Reduces collision risks involving vulnerable road users such as pedestrians and cyclists.
3. **Socially-Aware Decision Making**
   Integrates contextual reasoning, improving human-like driving behavior.
4. **Interpretability Through Chain-of-Thought Reasoning**
   Provides transparent reasoning outputs, supporting regulatory compliance and auditability.
5. **Edge-Compatible Deployment**
   Utilizes parameter-efficient fine-tuning techniques (LoRA) and small foundation models suitable for real-time embedded systems.

Overall, the system enhances both **technical robustness and commercial viability** by addressing a known limitation in current autonomous driving technologies.

6.3 Revenue Model

The commercialization strategy can adopt multiple revenue models:

1. Licensing Model

Autonomous vehicle manufacturers license the intent prediction module as a software component integrated into their AI stack.

2. Software-as-a-Service (SaaS) Model

Provide cloud-based training and model update services for fleet operators requiring continuous adaptation to new urban environments.

3. Tiered Integration Packages

Offer different levels of system capability:

- Basic occlusion-aware detection
- Advanced reasoning and explainability
- Full VLA-based adaptive prediction system

4. Research and Industry Collaboration Contracts

Partner with automotive OEMs and AI research labs for co-development and customization.

6.4 Cost Estimation

The primary cost components include:

1. **Research and Development Costs**
    - Model training infrastructure (GPU resources)
    - Simulation setup (CARLA-based testing)
    - Dataset acquisition and preprocessing
2. **Operational Costs**
    - Cloud computing resources for training and evaluation
    - Continuous model updates
    - System testing and validation
3. **Integration Costs**
    - Software engineering for integration into vehicle platforms
    - API and middleware development
4. **Compliance and Certification Costs**
    - Safety validation
    - Regulatory testing procedures

However, the use of parameter-efficient fine-tuning methods reduces training costs compared to full-scale large model retraining.

6.5 Pricing Strategy

The pricing strategy will be based on a **value-based pricing model**, reflecting the safety and risk mitigation benefits offered by the system.

Possible pricing approaches include:

- Per-vehicle licensing fee
- Annual subscription fee for fleet-based systems
- Enterprise integration contracts for automotive OEMs

Pricing tiers may be structured according to:

- Level of reasoning capability
- Prediction performance benchmarks
- Customization requirements

By emphasizing safety improvements and regulatory readiness, the pricing strategy aligns with high-value enterprise customers rather than consumer-level applications.

6.6 Competitive Advantage

The proposed system offers several competitive advantages over existing solutions:

1. **Occlusion-Aware Intent Prediction**
   Directly addresses a critical gap in current autonomous driving systems.
2. **Agentic AI-Based Reasoning Framework**
   Moves beyond reactive detection toward proactive, context-driven anticipation.
3. **Explainable Decision-Making**
   Chain-of-Thought reasoning enhances transparency and regulatory acceptance.
4. **Computational Efficiency**
   Use of Small Language Models and LoRA ensures deployability in real-time automotive environments.
5. **Scalability Across Urban Contexts**
   Combination of real-world datasets and synthetic simulation supports adaptable deployment.

Together, these factors position the proposed system as a differentiated solution within the autonomous driving AI ecosystem.

6.7 Intellectual Property Considerations

The development of the proposed Agentic AI-based adaptive intent prediction system involves several intellectual property (IP) considerations related to algorithm design, model architecture, software implementation, and dataset usage.

First, the core contribution of this research lies in the design of an occlusion-aware reasoning framework that integrates Vision-Language-Action architecture, Chain-of-Thought reasoning, and causal inference mechanisms. The specific integration methodology, reasoning workflow, and adaptive intent prediction pipeline may qualify for intellectual property protection, particularly in the form of a software patent or proprietary algorithm design, subject to institutional and jurisdictional regulations.

Second, the research utilizes publicly available datasets such as nuScenes and DriveLM, as well as simulation environments like CARLA. These resources are governed by specific licensing agreements. The project will strictly comply with all dataset usage policies, ensuring that no proprietary data is redistributed or commercially misused. The system will not claim ownership of third-party datasets or foundation models but will instead focus on novel architectural integration and algorithmic adaptation.

Third, the fine-tuning approach using Low-Rank Adaptation (LoRA) and the selected foundation model (e.g., Phi-3 Vision) will adhere to the respective open-source or research

licensing terms. Any derivative models developed as part of this research will clearly acknowledge the base model and respect its licensing constraints.

In a commercialization context, potential intellectual property strategies may include:

- Filing a patent for the occlusion-aware intent reasoning framework.
- Protecting source code under software copyright.
- Establishing licensing agreements for industry integration.
- Registering proprietary system architecture designs developed during the research.

Additionally, collaboration agreements with industry partners would require clear ownership terms regarding jointly developed enhancements or deployment adaptations.

By proactively addressing intellectual property considerations, this research ensures responsible innovation while preserving opportunities for commercialization and academic dissemination. The IP strategy balances open research contribution with protection of novel algorithmic advancements developed within this project.

7. BUDGET AND JUSTIFICATION

The successful development and evaluation of the proposed Agentic AI-based intent prediction system require allocation of resources across hardware, software, cloud infrastructure, and human effort. The following section outlines the estimated budget components and provides justification for each expense.

7.1 Hardware Costs

Item	Estimated Cost	Justification
External Storage (2TB SSD)	$150	Required for storing large-scale datasets such as nuScenes and synthetic simulation outputs from CARLA.

Hardware Justification

Training multimodal AI models and conducting simulation-based experiments require substantial computational resources. A GPU-enabled system ensures feasible model training and experimentation within academic time constraints. Local infrastructure also reduces long-term dependency on cloud services.

7.2 Software Licenses

Item	Estimated Cost	Justification
Development Tools (IDE / Professional Software)	$100	Optional licensed tools for debugging and model profiling if required.
Simulation Platform (CARLA)	$0	Open-source platform; no licensing fee required.
Machine Learning Frameworks (PyTorch, Hugging Face)	$0	Open-source frameworks available under permissive licenses.

Software Justification

The research primarily relies on open-source tools and frameworks to minimize costs while maintaining research reproducibility. Most required AI libraries are freely available, making the project economically efficient.

7.3 Cloud Services

Service	Estimated Cost (Project Duration)	Justification
Cloud GPU Training (AWS / Azure / GCP)	$600	Used for large-scale fine-tuning and performance benchmarking beyond local GPU capacity.
Cloud Storage and Backup	$150	Secure storage of trained models and experimental logs.
Experiment Tracking Platforms	$100	Tools for experiment reproducibility and performance comparison.

Cloud Services Justification

While local hardware supports primary experimentation, cloud GPU instances are necessary for training larger model variants and conducting controlled benchmarking. Cloud resources ensure scalability and reproducibility.

Human Effort Justification

The majority of development effort is contributed as part of academic research. However, expert consultation may require compensation to validate reasoning outputs and ensure domain-level reliability.

8. WORK BREAKDOWN STRUCTURE (WBS)

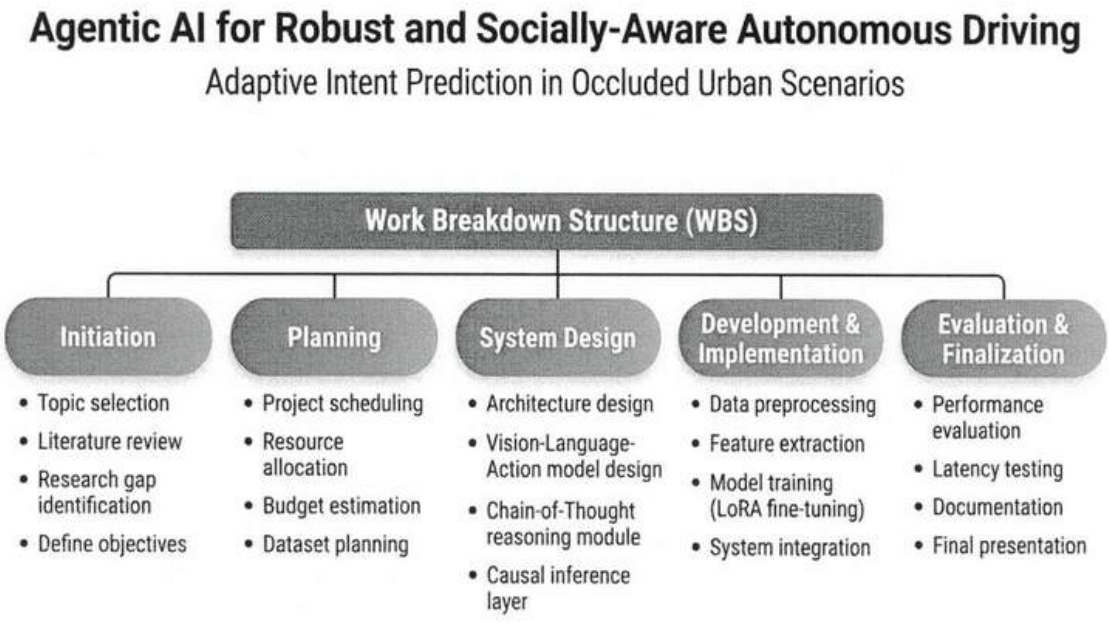


Figure 2 : work-breakdown

9. GANTT CHART

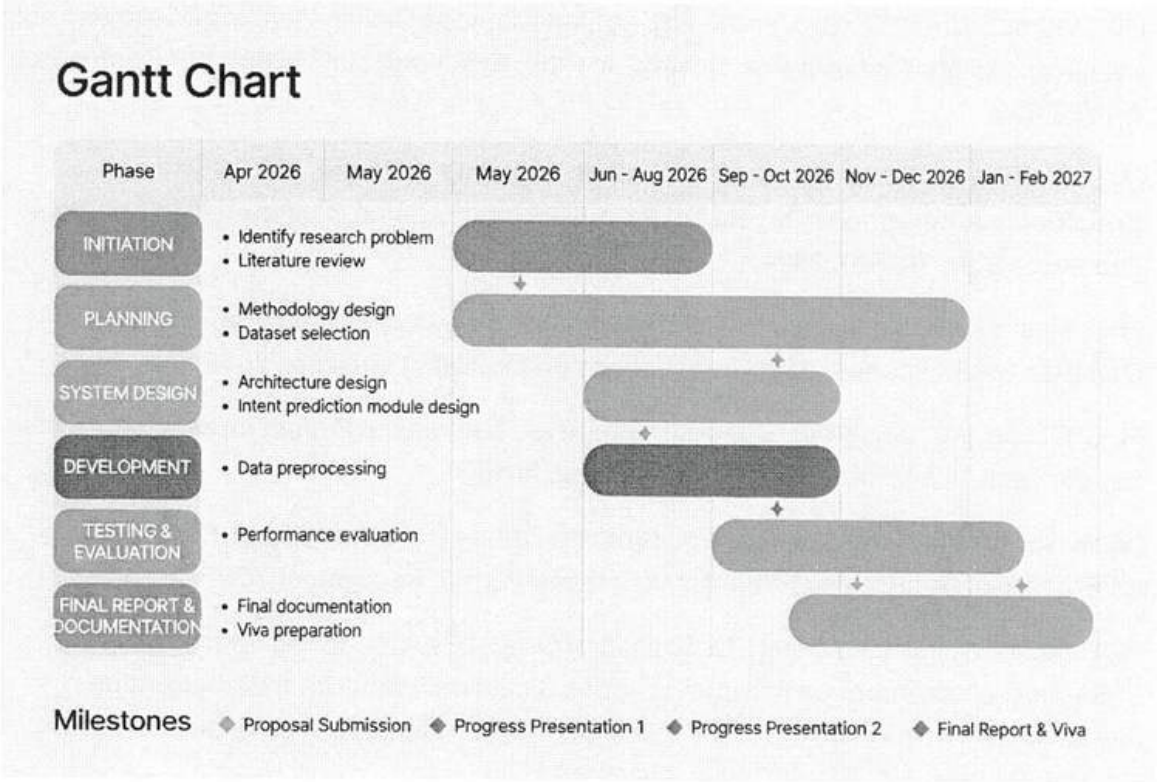


Figure 3 : Gantt chart

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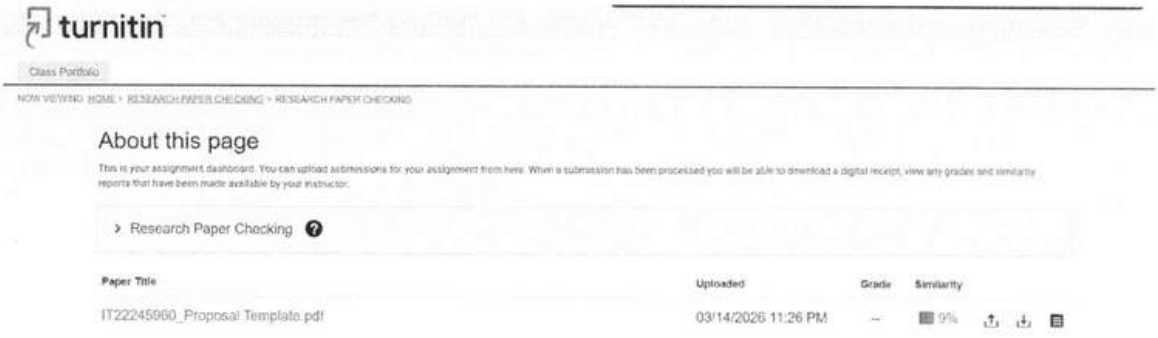
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11.APPENDICES

11.2 Turnitin Report



Equation 1: The report shows a similarity index of 9%